Definition 0.1 (Shannon Entropy): *Let P be a probability measure on a measurable space $(\mathcal{X}, \mathcal{F})$, and let $p = \frac{dP}{d\mu}$ be its density with respect to a reference measure μ. The Shannon Entropy $H(P)$ is defined as:*

$$H(P) = -\int_{\mathcal{X}} p(x) \log p(x) \, d\mu(x) = -E_P[\log p(X)]$$

where the integral is taken over the support of P.

Remark 0.2: *Shannon entropy is not an intrinsic functional of the probability measure P alone, but depends on the choice of reference measure μ. Different choices of μ lead to entropy values that differ by an additive constant.*

Remark 0.3: *In the discrete setting, taking μ to be the counting measure recovers the usual discrete entropy. In the continuous setting, taking μ to be the Lebesgue measure yields differential entropy, which may be negative and lacks invariance under coordinate transformations.*

Remark 0.4: *The expression $H(P) = -E_P[\log p(X)]$ is well-defined only when $\log p$ is P-integrable. If this condition fails, the entropy may be infinite or undefined.*

Definition 0.5 (Cross Entropy): *Let P and Q be probability measures on $(\mathcal{X}, \mathcal{F})$. If P is absolutely continuous with respect to a reference measure μ (with density p) and Q has density q, the Cross Entropy $H(P, Q)$ is:*

$$H(P, Q) = -\int_{\mathcal{X}} p(x) \log q(x) \, d\mu(x) = -E_P[\log q(X)]$$

Remark 0.6: *Cross entropy is not symmetric in its arguments and is finite only when $P \ll Q$. If $Q(x) = 0$ on a set of positive P-measure, then $H(P, Q) = \infty$.*

Remark 0.7: *Cross entropy decomposes as*

$$H(P, Q) = H(P) + D_{\mathrm{KL}}(P \| Q),$$

whenever both sides are well-defined. This identity makes explicit the role of cross entropy in statistical estimation and maximum likelihood methods.

Definition 0.8 (Joint Entropy): *Let $P_{X,Y}$ be a joint probability measure on the product space $(\mathcal{X} \times \mathcal{Y}, \mathcal{F} \otimes \mathcal{G})$ with joint density $p(x, y)$ relative to $\mu \times \nu$. The Joint Entropy $H(X, Y)$ is:*

$$H(X, Y) = -\int_{\mathcal{X}} \int_{\mathcal{Y}} p(x, y) \log p(x, y) \, d\mu(x) d\nu(y)$$

Remark 0.9: *Joint entropy is defined relative to the product reference measure $\mu \times \nu$. As with marginal entropy, its value depends on the choice of reference measures.*

Remark 0.10: *The joint entropy satisfies $H(X, Y) \leq H(X) + H(Y)$ whenever the entropies are finite, with equality if and only if X and Y are independent.*

Definition 0.11 (Conditional Entropy): *Given a joint measure $P_{X,Y}$, the Conditional Entropy $H(Y|X)$ is the expected entropy of the conditional distributions:*

$$H(Y|X) = -\int_{\mathcal{X}} \int_{\mathcal{Y}} p(x, y) \log p(y|x) \, d\mu(x) d\nu(y)$$

where $p(y|x) = \frac{p(x,y)}{p(x)}$ is the conditional density. It satisfies the relation $H(Y|X) = H(X,Y) - H(X)$.

Remark 0.12: *The conditional density $p(y|x)$ is defined only up to P_X-null sets. Accordingly, conditional entropy is defined as an expectation with respect to P_X, and pointwise values of $p(y|x)$ outside sets of full measure are irrelevant.*

Remark 0.13: *The identity $H(Y|X) = H(X,Y) - H(X)$ holds whenever the relevant entropies are finite. This equality is a consequence of the chain rule for Radon–Nikodym derivatives.*

Remark 0.14: *Unlike marginal entropy, conditional entropy is invariant under changes of the reference measure on $\mathcal{Y}$, provided the joint measure $P_{X,Y}$ is fixed.*